COURSE TITLE	**: MATERIAL TECHNOLOGY AND HEAT TREATMENT**
COURSE CODE	**: 4111**
COURSE CATEGORY	**: B**
SEMESTER	**: 4**
PERIODS PER WEEK	**: 5**
PERIODS PER SEMESTER	**: 75**
CREDITS	**: 5**

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Mechanical properties & Structure of material	21
2	Ferrous metals Alloy steel	18
3	Non-Ferrous metals & alloys Material Testing Methods Heat Treatment	18
4	Polymers & Polymerization Mechanisms Thermo Setting &Thermo Plastics	18
	TOTAL	75

COURSE OUTCOMES

On completion of the course, the student should be able to:

1. Understand the Structure and Properties of various Materials
2. Know the various properties of Iron and Steel and change of properties Alloy steel on the addition of other elements
3. Know the Need and uses of non-ferrous metals and alloys in Engineering application
4. Understand the material testing method like distructive and non-distructive methods.
5. Understand various Heat treatment process.
6. Know about Polymers

SPECIFIC OUTCOMES

MODULE – I

1.1.**0** **Understand the Structure and Properties of various Materials and their importance in Modern Technology**

- 1.1.1 Classification of Engineering Materials
- 1.1.2 Select material for a specific component

- 1.1.3 Define the mechanical properties of material such as tensile strength, compression strength, ductility malleability, hardness, brittleness, elasticity plasticity, impact strength, fatigue, creep resistance.
- 1.1.4 Identify the structure of materials - Unit cell, space lattice, BCC,FCC,&HCP structure, Atomic Packing factor
- 1.1.5 Transition from liquid to solid-crystallization
- 1.1.6 Effect of grain size on Mechanical properties
- 1.1.7 Deformation of metals
- 1.1.8 Elastic Deformation
- 1.1.9 Plastic Deformation
- 1.1.10 Mechanism of plastic deformation-slip mechanism & twinning mechanism
- 1.1.11 Comparison between Elastic deformation & plastic deformation
- 1.1.12 Describe work hardening
- 1.1.13 Describe Re-Crystallization

MODULE – 2

2.1.0 Know the various properties of Iron and Steel and change of properties Alloy steel on the addition of other elements

- 2.1.1 Identify the properties of pig iron
- 2.1.2 Identify the properties of wrought iron
- 2.1.3 Distinguish the various types of cast iron, such as white CI, molted CI gray CI and alloy CI
- 2.1.4 Describe the various steel making processes - Bessemer process,LD process, Open hearth process, Electric Furnace-(arc & induction type)
- 2.1.5 Identify the properties of steel
- 2.1.6 Iron-Iron Carbide Phase diagram
- 2.1.7 Alloy steel – Definition Effect of Alloying elements on steel
- 2..2 Properties & use of alloy steel such as Nickel steel, stainless steel, Spring steel, H.S.S, Tool steel
- 2.2.3 Designation of steel as per BIS standard.

MODULE – III

3.1. 0 Know the Need and uses of non-ferrous metals and alloys in Engineering application

- 3.1.1 Nonferrous metals such as Aluminium, Copper, Nickel, Lead
- 3.1.2 Non ferrous Alloys
- 3.1.3 Copper Alloy
- 3.1.4 Brasses
- 3.1.5 Bronzes

3.1.6 Aluminium alloys
3.1.7 Duralumin
3.1.8 Y-alloy
3.1.9 Hindalium
3.1.10 Bearing and anti friction alloy
3.1.11 Properties and uses of Admiralty Bronze, Phosphor Bronze, Lead bronze, White meat, Bhabit metal

3.2. 0 Analyze material testing methods

3.2.1 Destructive & non Destructive testing methods
3.2.2 Describe Tensile strength test
3.2.3 Describe Hardness test
3.2.4 Describe Impact strength test
3.2.5 Describe Fatigue Test
3.2.6 Creep Test with Creep curve
3.2.7 Describe non destructive testing methods
3.2.8 X-ray Test
3.2.9 r-ray Test
3.2.10 Magnetic particle Test
3.2.11 Ultrasonic test
3.2.12 Crystallographic Test- X ray diffraction Techniques(powder method)
3.2.13 Scanning Electron Microscope (SEM)

<u>MODULE – IV</u>

4.1.0 Understand the various process of Heat Treatment

4.1.1 Define heat treatment
4.1.2 Objective of heat treatment
4.1.3 Heat treatment processes
4.1.4 Surface Hardening processes- case hardening, nitrating, cyaniding, flame hardening, Induction Hardening
4.1.5 Heat Treatment of Tool steels
4.1.6 Time Temperature Transformation relation
4.1.7 T-T-T diagram – determination of T-T-T diagram and bainite and marten site transformation
4.1.8 Martempering
4.1.9 Austempering
4.1.10 Retained austenite

4.1.11 Sub-zero treatment

4.1.12 Heat treatment furnaces-muffle furnace, ear bottom funnace, batch type, continuous type, induction type furnaces

4.1.13 Defect in Heat treatment process

4.2.0 Polymers and polymerization mechanism

4.2.1 Definition of the polymer isation

4.2.2 Types of polymerization mechanisms- addition polymerization and condensation polymerization

4.3.0 Classify the types of plastics - Thermo plastics & thermosetting plastics

4.3.1 Important thermo plastics – PVC, CPVC, PTFE(teflon), Acrylic(PMMA),ABS, PC,LDPE,HDPE

4.3.2 Important thermo setting plastics-Phenolics, Bakelite, Epoxies, Silicones , polysters

CONTENT OUTLINE

MODULE– I

Classification of engineering materials- mechanical properties of materials- Tensile strength, Ductility, Malleability, Hardness, Brittleness, Elasticity, Plasticity, Impact strength ,Toughness, Stiffness, Fatigue, Creep Unit cell- space lattice- crystal structure-BCC,FCC, HCP structure.

Atomic packing factor-Transition from liquid to solid- Crystallization-Effect of cooling rate- Grain size- Effect of grain size on mechanical properties Deformation of metals- Elastic deformation and plastic deformation- Mechanism of plastic deformation- Slip mechanism and Twinning mechanism- Work hardening

MODULE – II

Ferrous metals and Alloys

Properties of Pig Iron – properties of wrought iron – production and properties of cast Iron – Classification- white CI, Molted CI, Gray CI and Alloy CI. Production and properties of steel – Bessemer processes LD process, open hearth processes – Electric furnace process. BIS' designation of steel Iron-Iron Carbide Phase diagram

Alloy Steel: - Definition – Effect of alloying elements on properties of steel -Nickel, Chromium, Manganese, Molybdenum Silicon, Vanadium and tungsten – Properties and use of alloy steels such as Nickel steels, Tool and Die steel, HSS Tool steel, Stainless steel, spring steels

MODULE – III

Need for the use of non-ferrous metals and alloys in engineering application
Non ferrous metals: Properties and use of Aluminium, copper, Lead, Tin, Zinc, Magnesium, Nickel
Non-ferrous alloys – Copper alloys- Brasses (Cartidge brass, Admirality brass, Muntz metals, Naval Brass), Bronzes (Phosphor bronze, Beryllium Bronze, Aluminium Bronze)- Aluminium alloys (duralumin, Y-alloy, Hindalium)-Bearing or anti friction alloys-properties aqnd uses of admiralty gun metal, phoshor bronze, lead bronze, white metal, babbit metal Destructive testing method-

1. Tensile Test
2. Hardness Test
3. Impact Test
4. Fatigue Test
5. Creep Test

Non-destructive testing methods

1. X-ray Test
2. r-ray Test
3. Magnetic particle test
4. Fluorescent penetrant test
5. Ultrasonic test
6. Crystallographic test-X-ray diffraction technique based on Bragg's law of diffraction (powder method)- Electron microscopy (Scanning Electron Microscope)

MODULE – IV

Heat Treatment – definition – Objectives of heat treatment– Heat treatment processes -annealing (full annealing ,Process annealing, isothermal annealing) – normalizing – hardening and tempering – surface hardening-(case hardening, nitriding, cyaniding,flame hardening, induction hardening)- Heat treatment of tool steel-Time temperature Transformation relation-TTT diagram-determination of TTT diagrams-bainite transformation- martensite transformation austempering- martempering Retained austenite Effect of retained austenite in hardness of steel- sub zero treatment Heat treatment furnaces- Defects in heat treatment

Polymers & Polymerization:

Definition of polymerization – polymerization Mechanismsan -Addition polymerisation – condensation polymerization –Definition of plastics – types – Thermo setting plastics – Thermo plastics – Examples of thermo plastics ,properties and uses (polythene,PVC, CPVC,PTFE-Tefflon, acrylics- PMMMA)-Thermosetting plastics -Bakelite, phenolics, epoxies,silicons, polysters

TEXT BOOKS

1. Material Science and Processes (Hajra Choudhari)
2. Material Science and Metallurgy (R.K.Rajput)
3.Engg.Materials and Metallurgy(R. Sreenivasan)

REFERENCE BOOKS

1.Material science (R.S. Khurmi)
2.Material and Metallurgy(V.K.Manchanda)
3. Manufacturing process of Engg.Material (Serope Kalpak Jian)